

Deep Generative Genre Remastering

From “coat-of-paint” transfer to structure-aware genre reconstruction

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CMPUT 414, Winter 2026

Problem surface timbre transfer sounds filtered, not re-composed.

Thesis deconstruct content vs style, then rebuild audio inside a target genre space.

🔗 Core pipeline

Lab 1: disentangle content and style

Lab 2: calibrate a 160D target vector

Lab 3: remaster with codec and diffusion branches

Lab 4: extend to coherent long-form audio

🌟 Demo claim

Short-form genre remastering already clears our main project gates; long-form coherence is measurable and improving.

Why This Problem Is Hard

🔊 1. Surface transfer is not enough

Earlier audio style transfer often changed timbre while leaving arrangement untouched. The result sounds filtered, not genuinely re-authored.

🧠 2. Music is entangled at multiple levels

Melody, rhythm, harmony, articulation, and texture are coupled. Without disentanglement, style edits also damage the melodic skeleton.

📖 3. What the literature points toward

Dai et al. identify the “coat-of-paint” problem; Hung and Yang show disentanglement; Roberts, Ho, and Meng motivate structure-aware generation with source anchoring.

✨ Our project hypothesis

Genre remastering should preserve structure, replace the style blueprint, and reconstruct with a generator that respects audio realism instead of just repainting the waveform.

Literature anchors: Dai et al. (2018), Hung et al. (2019), Yang et al. (2019), Roberts et al. (2018), Ho et al. (2020), Meng et al. (2022).

What We Built

🔧 Lab 1

Deconstruction encoder

GRL disentanglement, music gate

Style probe 0.942, leakage 0.108, gate AUC 0.930

📊 Lab 2

Target vector space

160D blueprint = z_{style} + spectral descriptors

Silhouette 0.494, probe acc 0.855

🔧 Lab 3

Reconstruction de-coder

EnCodec latent translator + diffusion branch

Best codec run: MPS 0.956, style conf 0.894

🔧 Lab 4

Long-form coherence

SDEdit anchoring, prefix lock, re-anchoring

160s output over 64 overlapping chunks

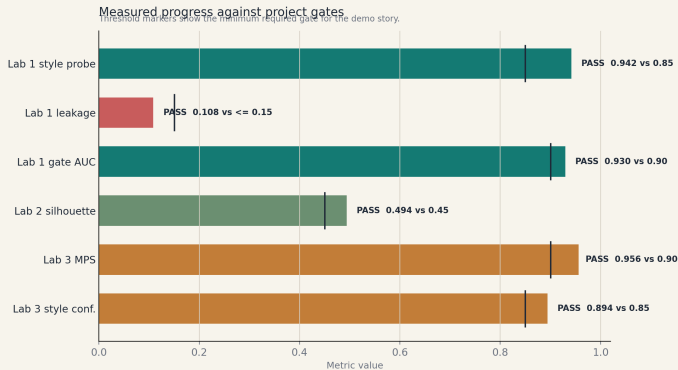
Engineering decisions that changed the results

1. EnCodec as a decoder guardrail to avoid raw-audio instability.
2. MERT-probe embeddings for cleaner style conditioning.
3. Direct-output translator to remove the residual style ceiling.

Progress by lab

Labs 1–4 are implemented in code, notebooks, and saved runs. Lab 5 already has listening-panel tooling, but the final blind study is still pending rather than part of our current evidence.

Measured Progress



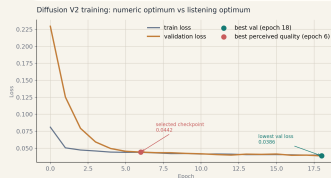
Main result

The project gates in Labs 1–3 are now measured wins, not just planned targets.

Best short-form run

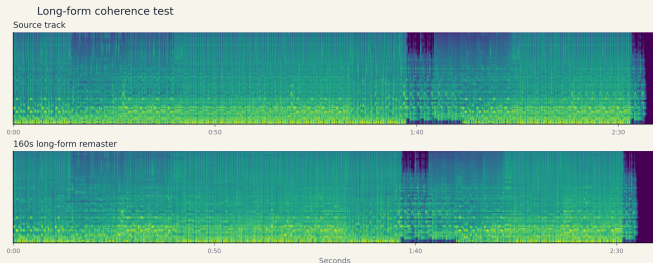
run1055 is the strongest demo candidate:

MPS = 0.956, style confidence = 0.894,
style accuracy = 0.949.



Interpretation: the codec branch is currently our most reliable demo path, while diffusion remains important because it gives stronger edit freedom and enabled the long-form experiments.

Long-Form Coherence and Demo Story



⚡ What Lab 4 shows

We can remaster a full 160-second track by stitching 64 overlapping chunks while keeping the structure visibly coherent over time.

Coherence metrics

boundary mel MSE mean: **0.00183**

boundary discontinuity mean: **2.87 dB**

Remaining issue: perceptual warble/static accumulation, not hard seams.

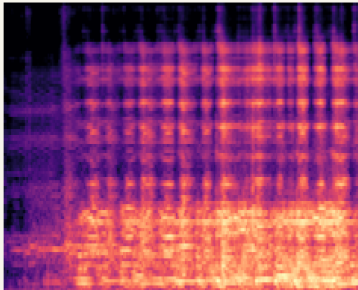
How we will present the demo

Show the DGGR pipeline, play 2 short codec remasters from run1055, then play 1 long-form sample and close with Lab 5 as the final human-validation layer.

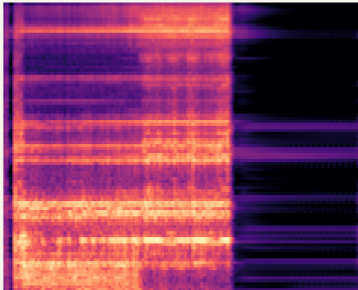
Backup: Example Short-Form Outputs

Best codec run (run1055): example remastered outputs

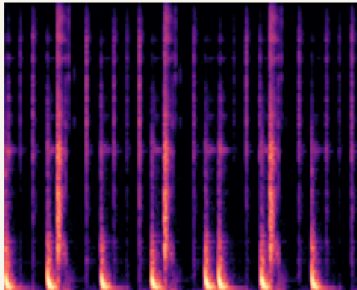
CC0/other -> Lo-fi



Hip-hop -> CC0/other



Lo-fi -> Baroque



All three clips are drawn from the shipped example set in `examples/audio/` and come from the best codec transfer run, run1055.

Backup: Sources and Visual Credits

Primary project references

Dai et al. (2018), Hung et al. (2019), Yang et al. (2019), Roberts et al. (2018), Donahue et al. (2019), Engel et al. (2019), Ho et al. (2020), Ho and Salimans (2021), Meng et al. (2022), Lee et al. (2023), Perez et al. (2018).

Repo-backed evidence used in these slides

Lab 1 audit summaries under `saves/lab1_run_combo_af_gate_exit_v2/`, Lab 2 validation summary under `saves/lab2_calibration/lab2_20260211_015118_lda_cleanup_v2/`, Lab 3 codec metrics under `saves2/lab3_codec_transfer/run1055/`, diffusion history under `saves2/lab3_diffusion/run_d002/`, and long-form metrics under `saves2/lab4_longform_coherence/fullsong_test/`.

Visual credits

OpenMoji icons via the local `openmoji` LaTeX package (CC BY-SA 4.0). Charts and spectrograms were generated locally from repo artifacts using `presentation/generate_figures.py`.